Solution

We know that $nC_r = \frac{nP_r}{r!}$

$$120 = \frac{720}{r!}$$

$$r! = \frac{720}{120} = 6 = 3!$$

$$\Longrightarrow r = 3$$

Example 2.28

If $15C_{3r} = 15C_{r+3}$, find r

Solution

$$15C_{3r} = 15C_{r+3}$$

Then by the property,

$nC_x = nC_y \Longrightarrow x + y = n$, we have

$$3r + r + 3 = 15$$

$$\Longrightarrow r = 3$$

Example 2.29

From a class of 32 students, 4 students are to be chosen for a competition. In how many ways can this be done?

Solution

The number of combination $= 32C_4$

$$= \frac{32!}{4!(32-4)!}$$

$$= \frac{32!}{4!(28)!}$$

Example 2.30

A question paper has two parts namely Part *A* and Part *B*. Each part contains 10 questions. If the student has to choose 8 from part *A* and 5 from part *B*, in how many ways can he choose the questions?

Solution

In part A, out of 10 questions 8 can be selected in $10C_8$ ways. In part B out of 10 questions 5 can be selected in $10C_5$. Therefore by multiplication principle the total number of selection is

$$10C_8 \times 10C_5 = 10C_2 \times 10C_5$$

$$= \frac{10 \times 9}{2 \times 1} \times \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2 \times 1}$$

$$= 11340$$

Example 2.31

A Cricket team of 11 players is to be formed from 16 players including 4 bowlers and 2 wicket-keepers. In how many different ways can a team be formed so that the team contains at least 3 bowlers and at least one wicket-keeper?

Solution

A Cricket team of 11 players can be formed in the following ways:

(i) 3 bowlers, 1 wicket keeper and 7 other players can be selected in

$$4C_3 \times 2C_1 \times 10C_7 \text{ ways}$$

$$4C_3 \times 2C_1 \times 10C_7 = 4C_1 \times 2C_1 \times 10C_3$$

$$= 960 \text{ ways}$$

(ii) 3 bowlers, 2 wicket keepers and 6 other players can be selected in

$$4C_1 \times 2C_2 \times 10C_6 \text{ ways}$$

$$4C_1 \times 2C_2 \times 10C_6 = 4C_1 \times 2C_2 \times 10C_4$$

$$= 840 \text{ ways}$$

(iii) 4 bowlers and 1 wicket keeper and 6 other players $4C_4 \times 2C_1 \times 10C_6$

$$4C_4 \times 2C_1 \times 10C_4 = 420 \text{ ways}$$

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